5-Day Multidomain Recommender Systems Challenge

Same methods – different domains – daily presentations

Fashion **Recipes** **Scholar** **Movies**

Project Tracks – 4 Domains, Same Pipeline

Group 1 – Fashion Recommender

- Recommend clothing items / outfits based on style, color, category, price.
- Think about aesthetics, seasonality, and mix-and-match outfits.

Group 2 – Recipe Recommender

- Recommend recipes based on ingredients, difficulty, cuisine type.
- Think dietary constraints, time-to-cook, and healthy vs comfort food.

Group 3 – Scholar / Paper Recommender

- Recommend papers based on abstract, topic, and research area.
- Think novelty, diversity of topics, and avoiding echo chambers.

Group 4 – Movie Recommender

- Recommend movies based on genres, tags, and textual descriptions.
- Think mood, diversity of genres, and avoiding repetitive suggestions.

Same task every day for all groups → only the DOMAIN is different.

Daily Workflow – Same Structure for Every Group

Every day, each group must:

1) Build – Implement the day's methods on their assigned domain.

2) Explore – Visualise, test, and interpret results.

3) Reflect – Identify failures, biases, and surprises.

4) Present – Give a short group presentation at the end of the day.

Daily presentations (all groups):

- What did we build today?
- What worked / what failed?
- What did we learn about our domain?
- What will we try tomorrow (out-of-the-box ideas)?

Day 1 – Data, Exploration & Embeddings

Core goal: understand your dataset and build the first embedding space.

Technical tasks

Data Prep

- Load the dataset from the notebook for your domain.
- Clean obvious issues (missing titles, weird values, duplicates).

Embeddings

- Build text embeddings using the provided sentence-transformer model.
- (If available) build image/feature embeddings for your items.

Exploration

- Create 2D and 3D UMAP visualisations of the item space.
- Explore clusters – name them, interpret them, find weird cases.

End-of-Day Presentation (each group)

1–2 slides: dataset overview + what makes this domain unique.

2–3 plots: 2D/3D embedding visualisations with labels or colours.

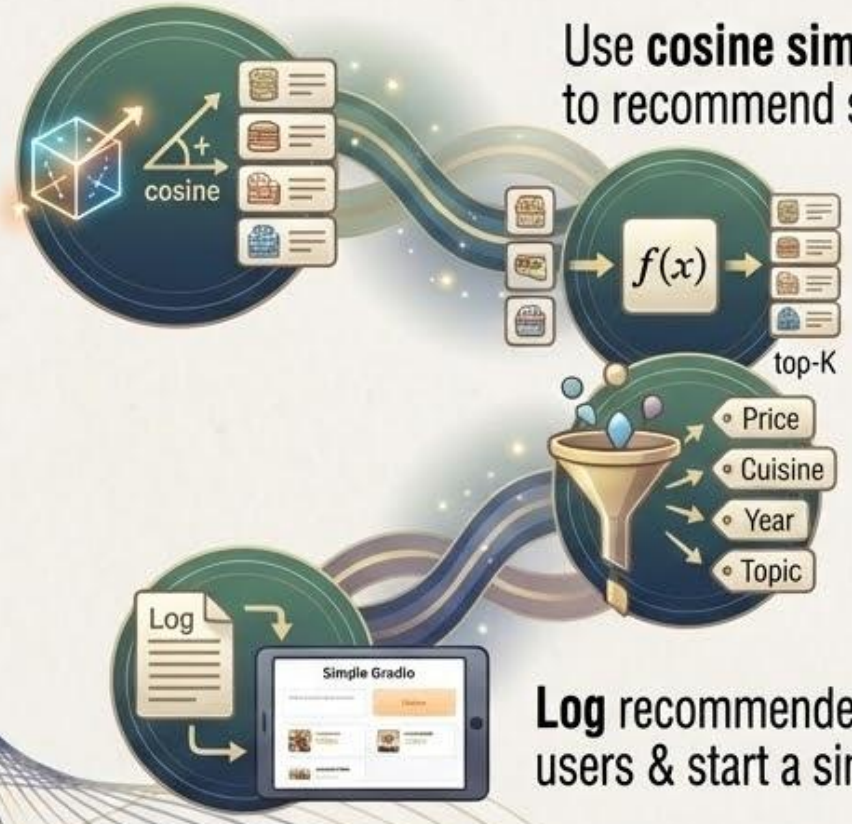
1 slide: interesting clusters / anomalies / biases you found.

1 slide: ideas for what a good recommender should optimise in this domain.

Day 2 – Basic Content-Based Recommender

Core goal: turn embeddings into the first working recommender.

Technical tasks



Use **cosine similarity** on embeddings to recommend similar items.

Build a simple function: given 1–3 liked items → recommend top–K.

Add simple **filters** (e.g., range, cuisine, year, topic).

Log recommended items for example users & start a simple Gradio prototype.

End-of-Day Presentation (each group)

Live demo: given a liked item, show top–K recommendations.

1 slide: how you defined "similarity" in your domain.

1 slide: qualitative evaluation – which examples look good / bad and why.

1 slide: first list of failure cases and how you might fix them tomorrow.

Day 3 – Personalisation & Hybrid Ideas

Core goal: move from item-to-item similarity to user-level personalisation.

All groups tasks

Build a **user embedding** from multiple liked items.

Let users express preferences (sliders, tags, constraints).

Combine similarity with simple constraints (budget, time, mood, etc.).

Experiment with re-ranking for diversity or serendipity.

Improve the Gradio app: personas, sliders, explanation text.

End-of-Day Presentation (each group)

Live demo: show a persona and how the recommendations adapt.

1 slide: how you modelled the "user" in your system.

1 slide: hybrid / constraint ideas you tried (even if they failed).

1 slide: how you make recommendations more diverse / less boring.

Day 4 – Final Gradio App & User Study Design

Core goal: finish the app and freeze the user study protocol.

All groups tasks

Finalise the Gradio interface (input flow, layout, explanation texts).

Ensure logging of: chosen options, recommended items, user ratings.

Decide what you want to measure: satisfaction, trust, diversity, etc.

Define your user study survey questions (SUS-like, 5-point Likert, etc.).

Test the full flow internally with classmates (pilot study).

End-of-Day Presentation (each group)

- Walkthrough of the final app flow in Gradio.
- **1 slide**: study design – target users, number of participants, questions.
- **1 slide**: which metrics you will compute from the logs.
- **1 slide**: hypotheses – what you expect to see and why.

Day 5 – Live User Study, Analytics & Competition

Core goal: run the study, analyse results, and compare systems.

During the lab session

All groups let users try their Gradio app (students cross-test).

Collect demographics: age, gender, nationality, study programme, etc.

Collect ratings for: relevance, satisfaction, trust, diversity, reuse intent.

Identify qualitative comments to quantitative metrics (e.g., Precision@K).

Export logs and compute your metrics in the notebook.

End-of-Day / Final Presentation (each group)

- **2–3 slides**: user demographics & participation stats.
- **2–3 slides**: recommendation quality metrics and visualisations.
- **1–2 slides**: bias / fairness observations.
- **1 slide**: your honest critique – where your system fails and how to improve it.

Winner Selection – Based on User Study Only

The winning team is determined purely by user study results.

Example combined score (each group computes this for THEIR system):

- Overall satisfaction (mean rating)
- Trust in the system
- Perceived relevance of recommendations
- Willingness to reuse / recommend to a friend

No extra points for fancy slides alone – the system must convince users.

Tie-breaker: team with more insightful analysis and clearer reflection on failures.